

K Subset Sums II

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Consider the sums of all $\binom{n}{m}$ subsets of the given array with exactly m elements.

Your task is to find the k smallest subset sums.

Input

The first line has three integers n , m and k : the size of the array, the size of the subsets and the number of subset sums k .

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print k integers: the k smallest subset sums in increasing order.

Constraints

- $1 \leq m < n \leq 2 \cdot 10^5$
- $1 \leq k \leq \min\left(\binom{n}{m}, 2 \cdot 10^5\right)$
- $-10^9 \leq x_i \leq 10^9$

Example

Input	Output
5 3 9 -3 1 5 2 0	-2 -1 0 2 3 3 4 6 7